# OsteoMap User Guide

**Instructions for Managing Specimens and Recording Skeletal Findings**

## 1. Introduction

OsteoMap is a desktop-style web application for recording and reviewing skeletal findings. It allows users to organise records by site, context, and specimen, annotate skeletal elements, view detailed anatomical regions, and import or export case data.

This guide explains how to create specimens, record skeletal observations, manage existing records, and back up data.

### Important: Data Storage

OsteoMap stores records locally in the browser using IndexedDB. Data is not automatically synchronised between devices. Clearing browser site data may remove locally stored records.

**Export regular backups** to reduce the risk of losing recorded information.

## 2. Home Page

The home page is the main dashboard for managing sites, contexts, and specimens.

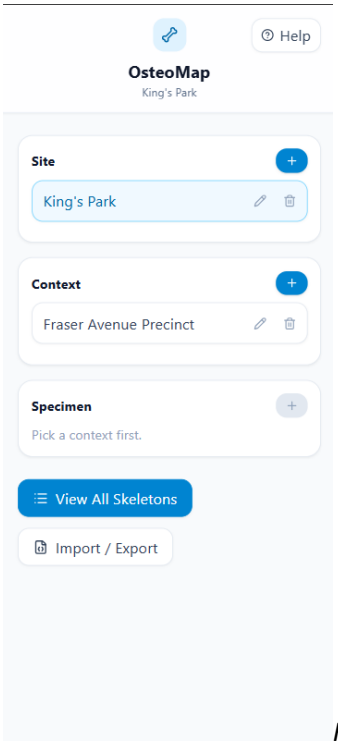



*Figure 1. OsteoMap home page showing the site, context, and specimen lists and the specimen preview panel.*

## 2.1 Understanding the Hierarchy

OsteoMap organises records into three levels:

- **Site:** The collection or location associated with a specimen.
- **Context:** A subgroup within a site, such as an excavation area or collection group.
- **Specimen:** An individual skeletal record.

Create a site first, then a context within that site, and finally a specimen within that context.

## 2.2 Adding a Site

1. Click **Add** in the Site column.
2. Enter the site name.
3. Save the new site.

The site should appear in the site list. Select it to manage its contexts.

## 2.3 Adding a Context

1. Select the relevant site.
2. Click **Add** in the Context column.
3. Enter a context name.
4. Save the context.

The context is associated with the selected site.

## 2.4 Adding a Specimen

1. Select a site and context.
2. Click **Add** in the Specimen column.
3. Enter a specimen ID.
4. Complete the available metadata fields, such as sex, age group, and notes.
5. Save the specimen.

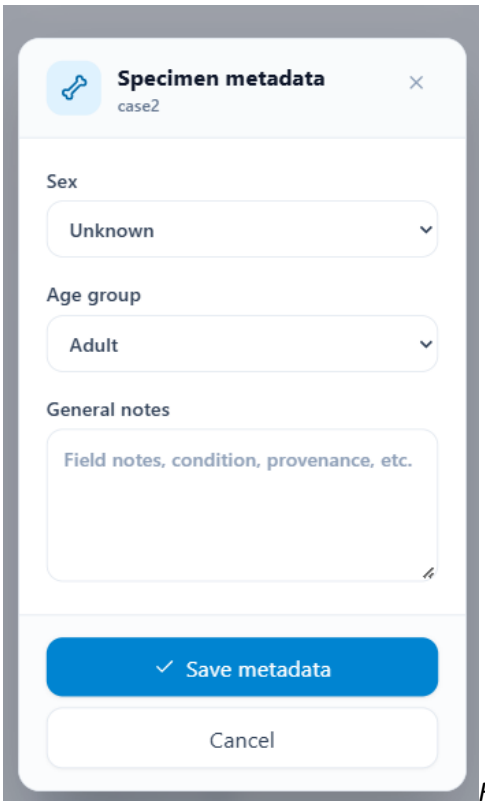


*Figure 2. Specimen creation dialog available metadata fields.*

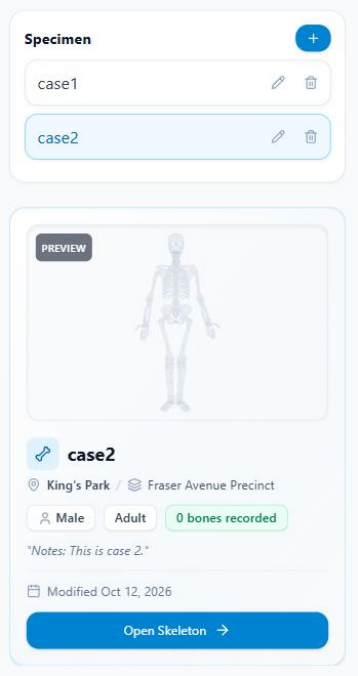
The new specimen should appear in the specimen list. Select it to view its details and open the skeletal recording page.

## 2.5 Specimen Preview

When a specimen is selected, the preview panel may display information such as:

- Specimen ID
- Associated site and context
- Sex and age group
- Recorded bone counts
- Last modified date
- Skeleton preview

Use the **Open Skeleton** button to access the skeletal recording page, if available.




# 3. Managing Specimens

## 3.1 My Skeletons Page

The My Skeletons page provides an overview of recorded specimens.

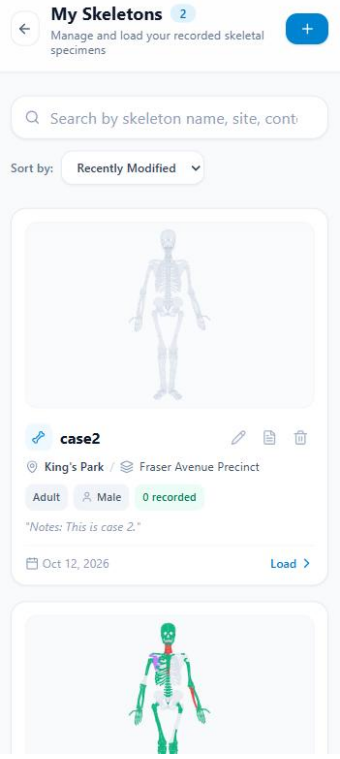



*Figure 3. My Skeletons page showing the specimen list*

Depending on the available controls, this page allows users to:

- Search for specimens.
- Sort records by name or modification date.
- Review specimen metadata.
- Open a specimen for further recording.
- Edit or delete existing records.

Search and filtering options may include specimen ID, site, context, sex, or age group.

### 3.2 Editing and Deleting Records

To edit a specimen, open its record and use the available metadata editing controls.

To delete a record, select the relevant delete option and confirm the action if prompted.

**Warning:** Deleting records may permanently remove data from the local database. Export a backup before deleting important records.

## 4. Recording Skeletal Findings

The skeletal recording page is the main workspace for documenting skeletal observations.

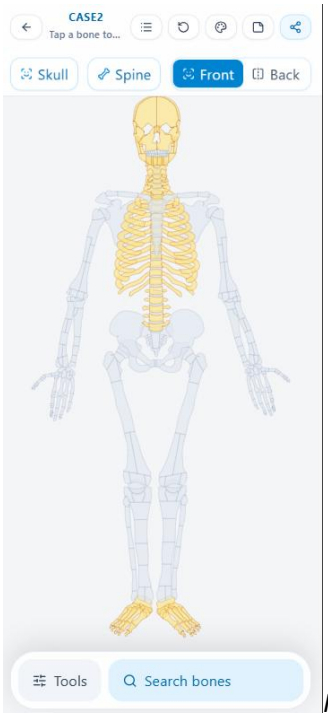



*Figure 4. Skeletal recording page showing the skeleton diagram, annotation toolbar, and navigation controls.*

## 4.1 Recording States

OsteoMap supports the following recording states:

- **Present:** The relevant skeletal element is recorded as present.
- **Absent:** The relevant skeletal element is recorded as absent.
- **Fragmentary:** The relevant skeletal element is recorded as fragmentary.

The application uses different colours to distinguish these states. Refer to the colour legend displayed in the application.

## 4.2 Marking Bones and Zones

The application provides different levels of anatomical selection:

- **Bone mode:** Selects an entire bone or anatomical structure.
- **Zone mode:** Selects a specific region or segment of a bone.

To record a finding:

1. Select the required recording state.
2. Choose bone mode or zone mode, as appropriate.
3. Select the relevant bone or anatomical region.
4. Confirm that the displayed state has changed as expected.

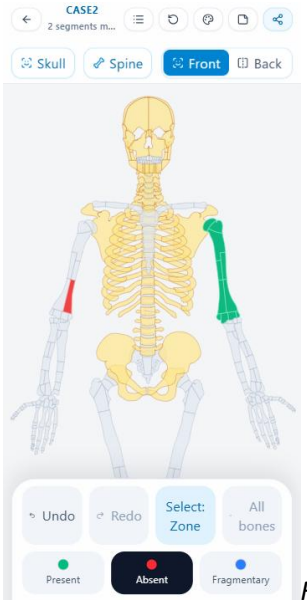



*Figure 5. Example of a skeletal element marked with a recording state.* On the left, the zone mode was used. On the right, the bone mode was used

Zone selection is useful when only part of a bone is present or when observations need to be recorded at a more detailed anatomical level.

## 4.3 Detailed Anatomical Views

The application may provide enlarged views of selected anatomical regions, allowing users to record findings more precisely.

Depending on the current implementation, these views may include:

- Skull
- Spine and ribs
- Left and right scapulae
- Left and right hip bones
- Left and right feet
- Individual vertebrae

Select the relevant region or detail-view control to inspect the available anatomy.

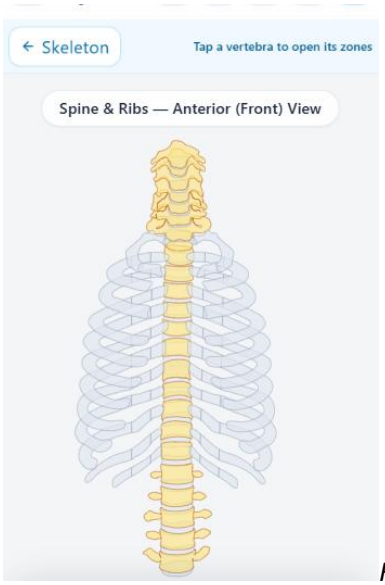



*Figure 6. Example of an enlarged anatomical view for more precise recording.*

## 4.4 Zoom and Navigation Controls

Use the available navigation controls to explore the skeleton. These may include:

- Zooming in and out.

- Switching between front and back views.
- Resetting the view.
- Opening detailed anatomical views.
- Searching for specific skeletal elements.

Use the controls shown in the application to navigate between the full skeleton and individual anatomical regions.

### 4.5 Notes and Observations

If the notes feature is available, use it to record relevant observations, such as:

- Condition of skeletal elements.
- Field observations.
- Contextual information.
- Interpretation comments.

Save any changes using the application's available controls.

## 5. Colour Legend

The colour legend explains the recording states displayed on the skeleton.

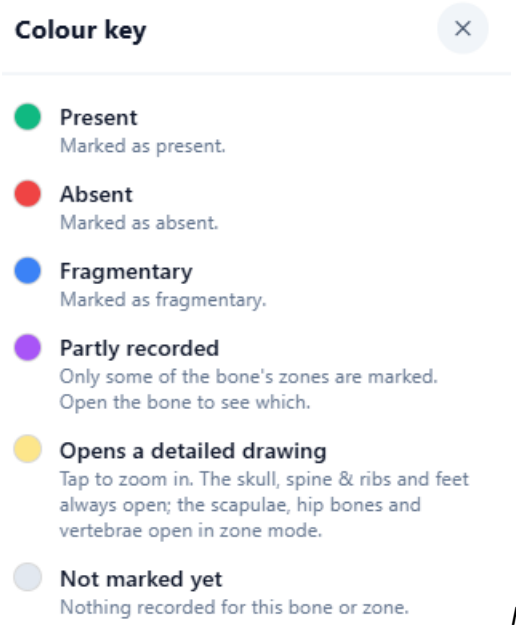



*Figure 7. OsteoMap colour legend.*

Use the actual legend in the application as the authoritative reference for colour meanings.

The expected recording states are:

- **Present:** Green.
- **Absent:** Red.
- **Fragmentary:** Blue.
- **Unmarked:** Neutral grey, if displayed.

Other visual indicators may identify partially recorded elements or regions that can be opened in a detailed view. Interpret these according to the application's displayed legend.

# 6. Importing and Exporting Data

OsteoMap provides import and export tools for backing up and transferring records.

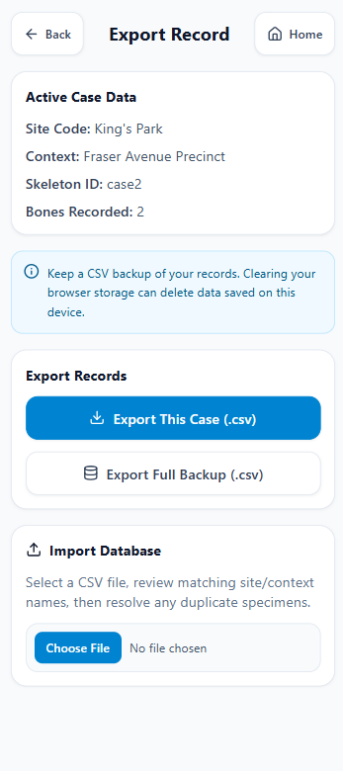



*Figure 8. Import and Export page showing the available data-management controls.*

## 6.1 Exporting a Specimen

If single-specimen export is supported:

1. Open the relevant specimen.
2. Navigate to the appropriate export control.

3. Select the option to export the current case.
4. Save the downloaded CSV file in a safe location.

## 6.2 Exporting a Full Backup

If full-database export is supported:

1. Open the Import / Export page.
2. Select the full-backup export option.
3. Save the downloaded file.

A full backup should preserve the records supported by the application's export format, including relevant site, context, and specimen information.

## 6.3 Importing Data

To import an existing CSV file:

1. Open the Import / Export page.
2. Select the CSV file to import.
3. Review the preview and any validation messages.
4. Review any duplicate or conflict warnings.
5. Resolve conflicts using the available options.
6. Confirm the import.

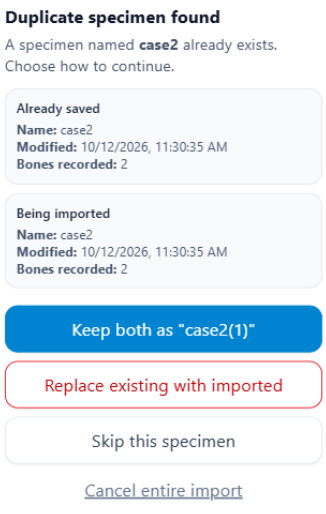



*Figure 9. Example of the import preview and any available conflict-resolution controls.*

After importing, check that the expected records are present and that their metadata and skeletal findings are correct.

# 7. Recommended Workflow

A typical recording session follows these steps:

1. Create or select a site.
2. Create or select a context.
3. Create or select a specimen.
4. Review the specimen metadata.
5. Open the skeletal recording page.
6. Mark the relevant bones or zones.
7. Review the colour-coded findings.
8. Add notes if supported and required.
9. Check the specimen record for completeness.
10. Export a backup when appropriate.

# 8. Best Practices

- Use consistent specimen IDs.
- Check the selected site, context, and specimen before recording findings.
- Use zone mode when detailed anatomical recording is required.
- Consult the colour legend when interpreting recorded states.
- Review records after making changes.
- Export regular backups.
- Verify imported records before relying on them for further work.

# 9. Troubleshooting

## 9.1 A Specimen Appears to Be Missing

- Check that the correct site and context are selected.

- Search for the specimen on the My Skeletons page.
- Confirm that you are using the same browser and device as before.
- Check whether browser site data has been cleared.
- Restore records from an available backup if necessary.

**9.2 A Specimen Cannot Be Created**

- Confirm that the required fields have been completed.
- Check that a site and context have been selected if required.
- Review any validation messages.
- Refresh the application and try again if the problem persists.

## 9.3 Changes Do Not Appear to Be Saved

- Check whether the application provides a save confirmation or other feedback.
- Reopen the specimen and verify the recorded information.
- Avoid clearing browser site data while records are stored locally.
- Export a backup once the data has been verified.

## 9.4 Importing a CSV File Fails

- Check that the file is in the expected CSV format.
- Review any error or validation messages.
- Confirm that the file has not been accidentally modified.
- Follow the application's duplicate-resolution instructions if conflicts are reported.

# 10. Developer Setup (Optional)

This section is intended for developers running OsteoMap from its source code, rather than users accessing an already running application.

## 10.1 Installing Dependencies

From the project root, run:

```
npm install
```

### 10.2 Starting the Application

Run:

```
npm run dev
```

Open the local URL printed by Vite in your browser.

Keep browser data intact and export backups regularly during development and testing.

## 11. Support

For problems that cannot be resolved using this guide, contact the project team and provide a description of the issue, the steps taken, and any relevant error messages.